

```
In [1]: # Print numbers from 1 to 10 using a for Loop.  
for i in range(1,11):  
    print(i)
```

1
2
3
4
5
6
7
8
9
10

```
In [2]: # Print even numbers between 1 to 50.  
for i in range(1,51):  
    if i%2==0:  
        print(i)
```

2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50

```
In [8]: # Print the sum of the first n natural numbers.  
sum = 0  
for i in range(int(input("enter vaule of n : "))+1):  
    sum = sum + i  
print(sum)
```

120

```
In [1]: # Print the multiplication table of a number.  
i = 1
```

```
n = int(input("enter n : "))
while i <= 10:
    print(n*i)
    i += 1
```

10
20
30
40
50
60
70
80
90
100

In [6]: *# Calculate the factorial of a number using a Loop.*

```
n = int(input("enter value of n : "))
fact = 1
for i in range(1,n+1):
    fact = fact * i
print(fact)
```

120

In [7]: *# Print all the digits of a number*

```
n = int(input("enter a number : "))
while n > 0:
    rem = n % 10
    print(rem)
    n = n // 10
```

3
2
1

In [8]: *# Count the number of digits in a number.*

```
n = int(input("enter a number : "))
count = 0
while n > 0:
    count += 1
    n = n // 10
print(count)
```

3

In [10]: *# Reverse a number using a Loop.*

```
n = int(input("enter a number : "))
rev = 0
while n > 0:
    rem = n % 10
    rev = rev * 10 + rem
    n = n // 10
print(rev)
```

321

```
In [12]: # Check if a number is a palindrome using a loop.
num = int(input("enter a number : "))
n = num
rev = 0
while n > 0:
    rem = n % 10
    rev = rev * 10 + rem
    n = n // 10
if num == rev:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Palindrome

```
In [15]: # Find the sum of digits of a number.
n = int(input("enter a number : "))
summ = 0
while n > 0:
    rem = n % 10
    summ = summ + rem
    n = n // 10
print(summ)
```

12

```
In [21]: # *
# * *
# * * *
# * * * *
for i in range(1,5):
    for j in range(0,i):
        print("* ",end="")
    print()
```

```
*
* *
* * *
* * * *
```

```
In [22]: # 1
# 1 2
# 1 2 3
# 1 2 3 4
for i in range(1,5):
    for j in range(0,i):
        print(j+1,end=" ")
    print()
```

```
1
1 2
1 2 3
1 2 3 4
```

```
In [26]: # A
# A B
```

```
# A B C
# A B C D
for i in range(1, 5):
    for j in range(65, 65 + i): # 65 is ASCII for 'A'
        print(chr(j), end=' ')
    print()
```

A
A B
A B C
A B C D

```
In [28]: # 1
# 2 2
# 3 3 3
# 4 4 4 4
for i in range(1,5):
    for j in range(0,i):
        print(i,end=" ")
    print()
```

1
2 2
3 3 3
4 4 4 4

```
In [29]: # * * * *
# * * *
# * *
# *
for i in range(1,5):
    for j in range(i,5):
        print("* ",end="")
    print()
```

* * * *
* * *
* *
*

```
In [37]: # Print all prime numbers between 1 to 100.
for num in range(2, 101):
    is_prime = True
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print(num, end=' ')
```

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97

```
In [38]: # Check if a number is prime using a loop
num = int(input("Enter a number: "))

if num < 2:
    print("Not a prime number")
else:
```

```

is_prime = True
for i in range(2, int(num**0.5) + 1):
    if num % i == 0:
        is_prime = False
        break
if is_prime:
    print("Prime number")
else:
    print("Not a prime number")

```

Not a prime number

```

In [39]: # Check if a number is an Armstrong number
num = int(input("Enter a number: "))
temp = num
sum_of_cubes = 0

while temp > 0:
    digit = temp % 10
    sum_of_cubes += digit ** 3
    temp //= 10

if sum_of_cubes == num:
    print("Armstrong number")
else:
    print("Not an Armstrong number")

```

Not an Armstrong number

```

In [40]: # Generate Fibonacci series up to n terms
n = int(input("Enter number of terms: "))

a, b = 0, 1
count = 0

while count < n:
    print(a, end=' ')
    a, b = b, a + b
    count += 1

```

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765

```

In [41]: # Find the Least Common Multiple (LCM) of two numbers
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    max_num = a
else:
    max_num = b

while True:
    if max_num % a == 0 and max_num % b == 0:
        print("LCM is:", max_num)
        break
    max_num += 1

```

LCM is: 64

In []: